

Index Printer Interface Protocol

For V5 & V4 Embossers

Index Braille © 2016

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Summary

This document applies to Index Braille V5 and V4 embossers.

Our Braille printers (embossers) have been on the market since 1983, so the communication interface is quite old fashion. Even today a lot of our customers are running DOS (believe it or not). The main goal for the new system is to make it easy for blind customers to use most of the applications on the windows platform that is by default accessible for sighted people.

Our goal is to spread information on how to make use of our embossers to other Braille software developers and hopefully this would evolve better editors and graphical tools for visually impaired.

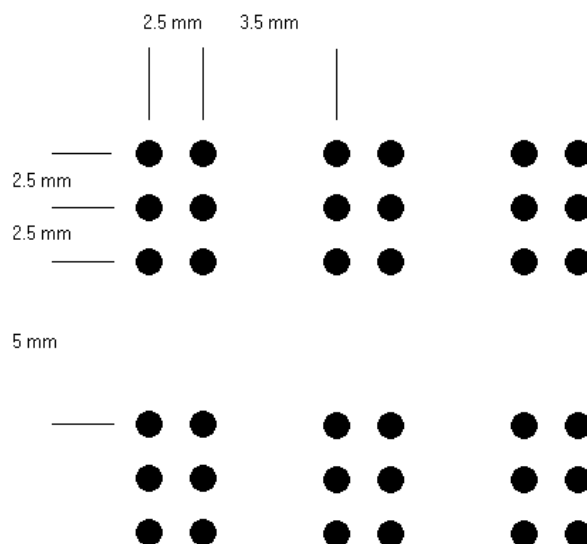
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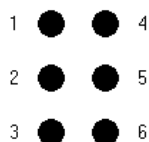
1 Introduction to Braille

The Braille system, invented by the blind Frenchman Louis Braille, became accepted in France in the 1850s. It has since then been widely spread around the globe, and is now being used by the blind community in most parts of the world as the major system for reading and writing.

The Braille alphabet consists of 63 characters, each containing up to six raised bulging dots individually separated with 2.5 mm, and with a fixed size of six mm in width and ten mm in height:



To aid in describing the characters, the six dots of a Braille cell are numbered 1, 2, and 3, downward on the left, and 4, 5, and 6, downward on the right, thus:



The 63 Braille characters are universally grouped in a table of seven lines:

1 st Line:									
2 nd Line:									
3 rd Line:									
4 th Line:									
5 th Line:									
6 th Line:									
7 th Line:									

The first line's characters comprise the four upper dots. The second, third and fourth lines add dot numbers three and six to the first line's dots. Line five is the same as the first line, but the dots are shifted to the four lower positions.

1.1 Interpretation of the Braille symbols

Unfortunately, there is no international standard on how to interpret the Braille characters. Every language has its own interpretation. To make it even worse, more than one interpretation exist in many languages.

The limited number of Braille characters makes it difficult to read capitals and small letters, numbers, and all punctuation symbols. The capitals, A to Z, the small letters, a to z, and the numbers, 0 to 9, are all together 62 to the number. If we add the punctuation symbols, we will far exceed the number of the 63 available Braille characters. Some Braille characters have therefore been set aside and used as prefix symbols to mark for example capitals and numbers, thus the letter “a”, the letter “A” and the number “1” all use the same Braille character ⠁. To distinguish between the three characters, the number is proceeded with the prefix ⠼, while the capital “A” is proceeded with the Braille character ⠠. This is valid in English, though. To further complicate the situation, the prefix symbols are not standardized. The French capital prefix is for example h. To summarize: The three characters “a”, “A” and “1” are typed in English Braille:

a: ⠁
A: ⠠⠁
1: ⠼⠁

Since Louis Braille was born in France, it is quit natural that the French interpretation of the Braille characters is close to Louis Braille's original interpretation:

1 st Line:	a	b	c	d	e	f	g	h	i	j
2 nd Line:	k	l	m	n	o	p	q	r	s	t
3 rd Line:	u	v	x	y	z	ç	é	à	è	ù
4 th Line:	â	ê	î	ô	û	ë	ï	ü	œ	w
5 th Line:	,	;	:	/	?	!	"	(	*	)
6 th Line:	3	§	1	æ	'	—				
7 th Line:	3	3	3	3	2	3	3			

¹ Number prefix

² Capital prefix

³ Modification prefixes for character variants and formats.

Note: The letter “w” was not part of the original interpretation.

None of the Braille characters are interpreted as a number, as discussed above. To emboss a number, we use the letters “a” to “j” in combination with the number prefix symbol ⠼.

1 :		6 :	
2 :		7 :	
3 :		8 :	
4 :		9 :	
5 :		0 :	

Only one number prefix is needed in front of a number. Example: The number 2002 is embossed in Braille:



Many of the French Braille symbols cannot be expressed in written text, since they are used as modification prefixes, and thus have no printable character counterpart. Most of them are located in the sixth and the seventh line. Two interpretations exist, though, where all Braille characters are interpreted as printable text letters: the American and the European interpretations. The American interpretation is sometimes referred as the MIT Braille, since it was invented at MIT, the Massachusetts Institute of Technology. The European interpretation is known as Euro Braille or the IBM437 code. The MIT Braille interpretation is as follows:

1 st Line:	a	b	c	d	e	f	g	h	i	j
2 nd Line:	k	l	m	n	o	p	q	r	s	t
3 rd Line:	u	v	x	y	z	&	=	(	!	)
4 th Line:	*	<	‰	?	:	\$	]	\	[	w
5 th Line:	1	2	3	4	5	6	7	8	9	0
6 th Line:	/	+	#	>	'	-				
7 th Line:	@	^	_	"	.	;	,			

The MIT Braille interpretation is used for most languages. A few European languages, notably German, use the Euro Braille interpretation, as follows:

1 st Line:	a	b	c	d	e	f	g	h	i	j
2 nd Line:	k	l	m	n	o	p	q	r	s	t
3 rd Line:	u	v	x	y	z	&	%	[	^	]
4 th Line:	1	2	3	4	5	6	7	8	9	w
5 th Line:	,	;	:	/	?	+	=	(	*	)
6 th Line:	\	0	#	@	.	-				
7 th Line:	"	>	_	!	\$	<	'			

The clear-sighted might recognize that the umlaut-characters (ä, ö and ü) are missing in the Euro Braille table. How can the German language use the Euro Braille interpretation, then? Don't we need all characters to be able to read proper German? That is true. The German letter "ä" is shown in Braille as ⠏⠗, the letter "ö" is ⠕⠗, and "ü" is ⠥⠗. But, as you probably already have noted, the Braille symbol ⠏⠗ represents the at-sign "@", the symbol ⠕⠗ represents "9" and the symbol ⠥⠗ represents "8". Though, the Braille reader will read the letters ä, ö and ü when he reads the Braille symbols ⠏⠗, ⠕⠗ and ⠥⠗, just because he knows he is reading German. But what happens if we want to emboss the "@", "8" or "9"? The numbers are not a problem. All numbers are embossed using the prefix | and the letters "a" to "j". The at-sign, on the other hand, is embossed using the symbols ⠏⠗⠐, i.e., we simply add the Braille symbol ⠐, with a single dot four, to tell the reader that he is reading the "@"-sign and not the letter "a".

This is complicated! Can we really say that the German language uses the Euro Braille table? Well. This is a matter of definition. The reason why we still keep saying that will be clear, or at least clearer, when we look at how the Braille reader uses the computer with a screen reader (a program that “speaks” the text on the screen through the computer’s loudspeakers) and a Braille display.

1.2 Contracted Braille

As another example we can look at the interpretation of the Braille characters in the English language. Let us look at the first four lines. With the Braille characters ordered as in the former examples, we have:

1st Line:

2nd Line:

3rd Line:

4th Line:

And the interpretation of the dots:

1st Line: a b c d e f g h i j
 2nd Line: k l m n o p q r s t
 3rd Line: u v x y z and for of the with
 4th Line: ch gh sh th wh ed er ou ow W

What is this? Some Braille characters are interpreted as words or syllables! This is quite convenient, since the Braille symbols are much more space consuming than the printed text characters. We can simply contract some often-used words into one single Braille symbol to save space. This is what is often called contracted Braille. Contracted Braille can be graded, depending on the number of available contractions. English Grade 1 Braille consists of the letters of the alphabet, punctuation, numbers and a few prefix symbols, like number prefix and capital prefix. Grade 1 Braille does not contain any word contractions. Grade 2 Braille consists of Grade 1 and 189 contractions and short-form words. When we speak about English Braille, we often mean English Grade 2 Braille. Contracted Braille exists in most languages. The contractions are of course different in different languages. The level of contractions differs also. German Grade 1 contains for example some few contractions, while the English Grade 1 does not.

Which level of contractions should we use? That depends on the reader. If the text or document is written in English for an English speaking audience, we will probably prefer the Grade 2 level. If the audience is international, with, presumable, less knowledge in English Braille, we will preferably use the Grade 1 Braille.

Braille contractions exist in a variety of forms in different languages, for example as one-cell signs for words and syllables as in the table above, as two-cell signs, or as short-form words with different kinds of prefixes.

2 The printer driver for Windows Vista/7/8/8.1/10

2.1 Scenarios

We can define two different scenarios of how to print to the driver.

2.1.1 Scenario 1: No escape sequences

All settings are done on the embosser and the input is assumed to be ready formatted with characters encoded according to embosser settings (you are supposed to have chosen the correct Braille table).

2.1.2 Scenario 2: Controlling the embosser via escape sequences

By using escape sequences in the beginning of a file one can control the embosser settings.

2.2 Page Layout of V5/V4 Embossers

2.2.1 Basic-D V5/V4



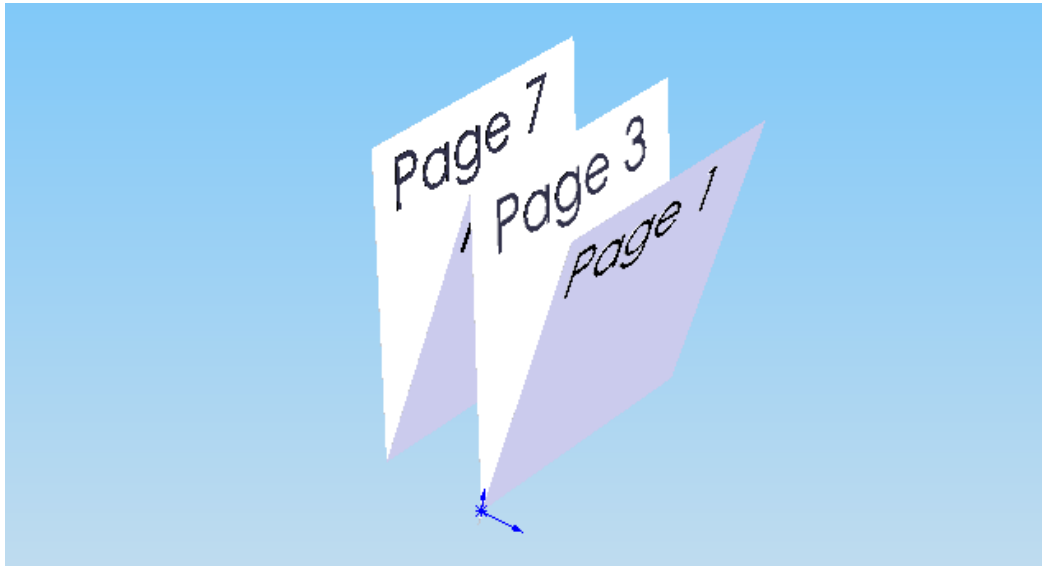
Maximum number of characters per line with standard 2.5 Braille is.....: 48
Maximum printable width.....: 301.152 mm
Maximum paper width.....: 325 mm

2.2.1.1 Unprintable area

The unprintable area is 5 mm on each side of selected paper. E.g. the printable area of an A4 paper is 200 mm and not 210 mm even though the embosser can handle 301.152 mm width paper.

2.2.1.2 Special printing modes

2.2.1.2.1 Z-Folding



The Basic-D V5/V4 supports the Z-folding mode. Z-folding means that the Basic-D V5/V4 prints the pages in such a way that when the printed document is finished, the user could take the paper bundle, cut it and directly bind it without reorganize the pages.

2.2.1.2.2 Z-Folding sideways

This is a new format where the printout is rotated 90°. With this format only one side has to be cut before binding.

2.2.1.2.3 Double sided/Single sided

The Basic-D V5/V4 supports single sided and double sided printout.

2.2.2 Everest-D V5/V4



Maximum number of characters per line with standard 2.5 Braille is: 47
Maximum printable width..... : 297 mm
Maximum paper width.....: 297 mm

2.2.2.1 Unprintable area

The unprintable area is 5 mm on each side of selected paper. E.g. the printable area of an A4 paper is 200 mm and not 210 mm even though the embosser can handle 301.152 mm width paper.

2.2.2.2 Special printing modes

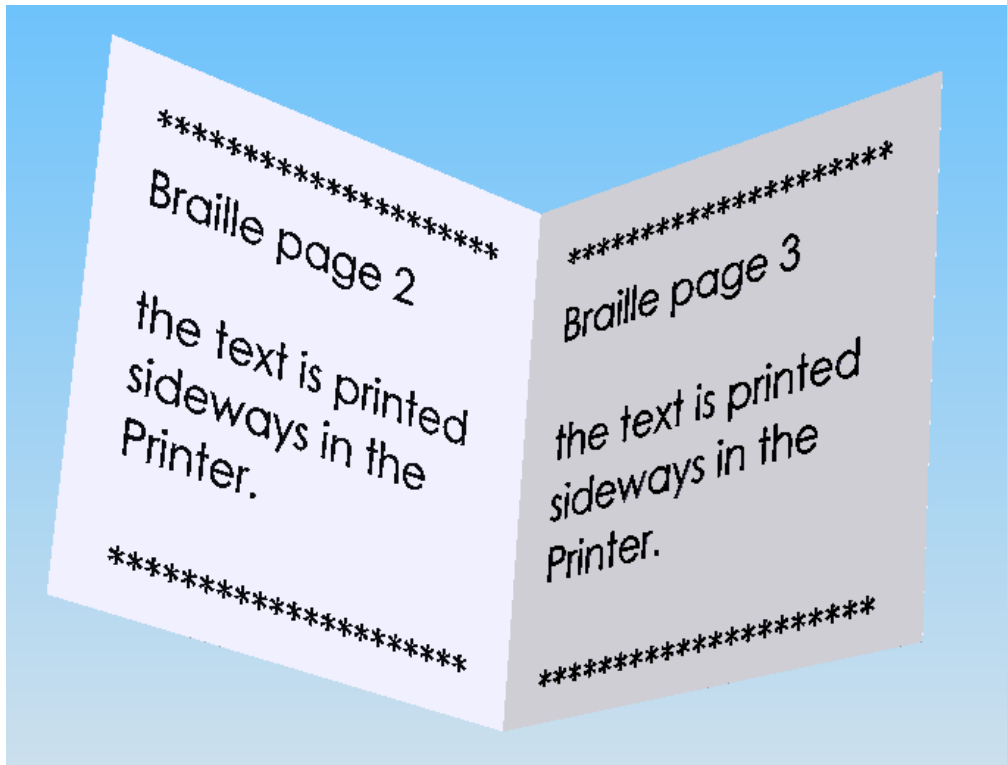
2.2.2.2.1 *Double sided/Single sided*

The Everest-D V5/V4 supports single sided and double sided printout.

2.2.2.2.2 *Saddle-Stitch Binding*



The Everest-D V5/V4 supports saddle-stitch binding. This printing mode enables the user to produce own magazines.



If printing in saddle-stitch mode, the printer is loaded with A3 paper, the printer expects the page length and page with parameters to be A4 size, not paper size A3.
 Everest-D V5/V4 supports Jumbo Braille and Japanese 2.2 Braille in this mode.

2.2.3 BrailleBox V5/V4



Maximum number of characters per line with standard 2.5 Braille is.....: 47
 Maximum printable width.....: 297 mm
 Maximum paper width.....: 297mm

2.2.3.1 Unprintable area

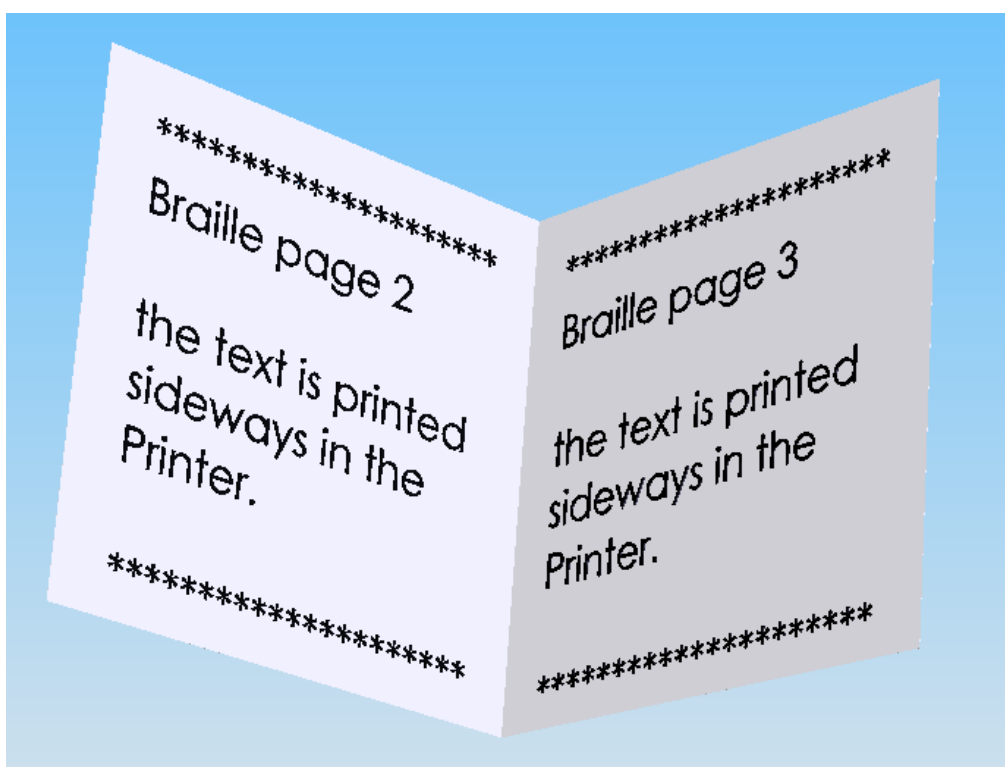
The unprintable area is 5 mm on each side of selected paper. E.g. the printable area of an A4 paper is 200 mm and not 210 mm even though the embosser can handle 301.152 mm width paper.

2.2.3.2 Special printing modes

2.2.3.2.1 Saddle-Stitch Binding



The Braille Box V5/V4 supports saddle-stitch binding. This printing mode enables the user to produce own magazines.



If printing in saddle-stitch mode, the printer is loaded with A3 paper, the printer expects the page length and page with parameters to be A4 size, not paper size A3.
Unlike Everest-D V5/V4 the BrailleBox V5/V4 doesn't support Jumbo Braille and Japanese Braille in this mode.

3 Implemented escape sequences in V5/V4 embossers

The Index Braille V5/V4 embossers are not completely compatible with all ESC sequences that were implemented in V1, V2 and V3 embossers. Below is a table with all ESC sequences that is implemented in V5/V4 embossers.

Decimal	Hexadecimal	Description	Reference
27 1 [data]	1B 01	Activate six dot graphics	See 3.1
27 2	1B 02	Deactivate six dot graphics	See 3.1
27 3 [data]	1B 03	Activate general graphics	See 3.2
27 4	1B 04	Deactivate general graphics	See 3.2
27 6	1B 06	Deactivate four dot graphics	See 3.3
27 7 [data]	1B 07	Activate four dot graphics	See 3.3
27 9	1B 09	Activate image graphics	See 3.4
27 10	1B 0A	Deactivate image graphics	See 3.4
27 11 r h v ls	1B 0B r h v ls	Set image data	See 3.5
27 49	1B 31	Activate six dot graphics	See 3.1
27 68	1B 44	Activate temporary formatting properties of a document.	See 3.6
27 92 x y [data] 13 10	1B 5C x y [data] 0D 0A	Transparent mode, multiple characters	See 3.7
27 93	1B 5D	Deactivate six dot graphics	See 3.1
27 94 c	1B 5E c	Transparent mode, one character	See 3.8
27 252	1B FC	Activate / deactivate Letter graphic	See 3.9

Table 1: Implemented ESC codes in Index Braille V5/V4 embossers

3.1 Activate / deactivate six dot graphic

In this mode, the embosser uses Index Braille table and translates each character to a 2x3 graphics matrix that is identical to corresponding Braille cell.

Syntax:

27 01 [data] 27 02 or

27 49 [data] 27 93

[data] = an array of the graphics data in the form of bytes.

3.2 Activate / deactivate general graphic

In this mode, the embosser uses the current Braille code and translates each character to a 2x3 or 2x4 graphic matrix that is identical to corresponding Braille cell. The size of the matrix is dependent on the number of dots that is defined in the Braille code information.

Syntax:

27 3 [data] 27 4

[data] = an array of the graphics data in the form of bytes.

3.3 Activate / deactivate four-dot graphic

In this mode, the embosser uses only the four less significant bits in the eight-bit computer code to create Braille cells. Each character is translated to a 1x4 dot matrix according to the table below:

Bit1
Bit2
Bit3
Bit4

Two character cells are needed to create one 2x4-dots Braille cell. First character will give the first column and the second character will give the second column. Table 1: Part of ASCII table that can be used for four dot graphics

defines a part of the ASCII table and can be used to derive the dot combination for some characters.

Syntax:

27 7 [data] 27 6

[data] = an array of the graphics data in the form of bytes.

Decimal	Hexadecimal	ASCII	Less significant
64	40	@	0000
65	41	A	0001
66	42	B	0010
67	43	C	0011
68	44	D	0100
69	45	E	0101
70	46	F	0110
71	47	G	0111
72	48	H	1000
73	49	I	1001
74	4A	J	1010
75	4B	K	1011
76	4C	L	1100
77	4D	M	1101
78	4E	N	1110
79	4F	O	1111

Table 1: Part of ASCII table that can be used for four dot graphics

3.4 Activate / deactivate image graphic

This parameter is for image graphics. It is described in detail in the technical example: 3.10 Graphics example for V5/V4 embossers.

3.5 Set image data

This parameter is for image graphics. It is described in detail in the technical example: 3.10 Graphics example for V5/V4 embossers.

3.6 Temporary settings of document properties

The formatting properties for a document in the Index Braille embossers are set through layout 1 to 5 (in V3 embossers this was called setup A, B and C). These settings are only applied to plain text document.

The V4 and V5 models of Index Braille embossers have the capability to temporary change some formatting properties for a document. This capability is implemented in firmware version 10.30 and higher. The properties that can be changed are specified in table 2.

Property	Parameter	Comment	Reference
Binding Margin	BI	V4 and V5 embossers. Not V3.	See 3.6.2
Braille Table	BT	From version 11.02.1.	See 3.6.3
Characters per Line	CH		See 3.6.4
Page Mode(duplexing)	DP		See 3.6.5
Driver Version	DV		See 3.6.6
First Line Offset	FO		See 3.6.7
Graphic Dot Distance	GD		See 3.6.8
Lines per page	LP	V4 and V5 embossers only.	See 3.6.9
Line Spacing	LS		See 3.6.10
Multiple Copies	MC		See 3.6.11
Multiple Impact	MI	From version 11.03.2	See 3.6.12
Page Number	PN		See 3.6.13
Text Dot Distance / Braille Cell Size	TD	From V4 it's called Braille Cell Size, but the parameter is same - TD.	See 3.6.14
Top Margin	TM		See 3.6.15
Paper Size	PA	Only V5 embossers, firmware 2.x	See 3.6.16

Table 2: Formatting properties that can be changed temporarily.

Syntax for changing properties

To change properties for a document there is a special escape sequence that should be added at the beginning of the document or beginning of a row. The escape sequence has a special syntax that must be followed. The syntax is *[ESC]DXX999, ... , ...;*

Description of syntax:

- [ESC]D - Start of temporary document properties.
- XX - Parameter to change, see description above
- 999 - New value for parameter, see description below.
- ,
- Separator between parameters
- ...
- Next parameter and value to change.
- ;
- Last parameter should end with semicolon to indicate end of changing properties.

For example to emboss three copies single sided with a top margin of two lines and bottom margin 1 of the document the syntax will be:

Example:

ASCII => [ESC]DMC3,DP1,TM2,LP27,BI1,CH30;

Decimal => 27 68 77 67 51 44 68 80 49 44 84 77 50 44 76 80 50 55 44 66 73 49 44 67 72 51 48 59

Hexadecimal => 1B 44 4D 43 33 2C 44 50 31 2C 54 4D 32 2C 4C 50 32 37 2C 42 49 31 2C 43 48 33 30 3B

Differences in V3 and V5/V4 page mode parameters

The difference in page layout between V3 and V5/V4 embossers are shown in Figure 1.

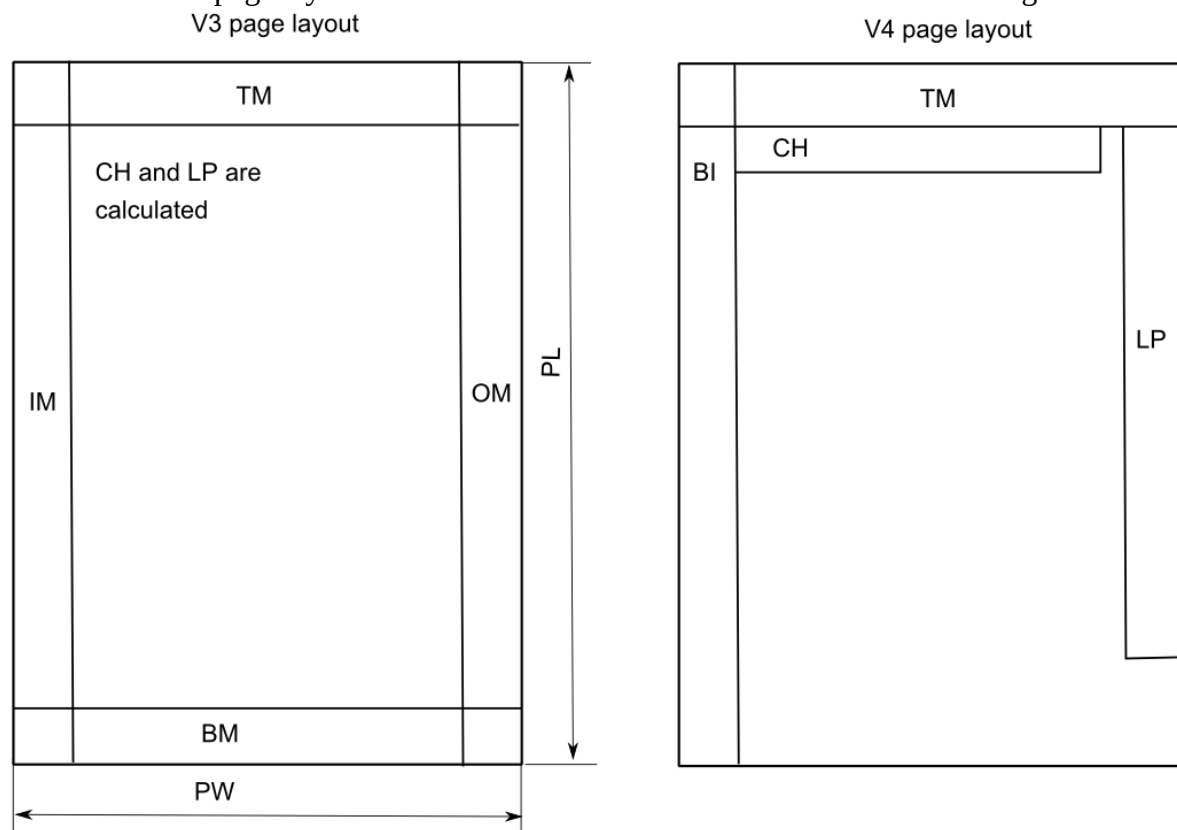


Figure 1

3.6.1.1 Example of parameters for V5/V4 embossers

[Example:

ASCII => ESC]DBI2,TM1,CH40,LP27;

Decimal => 69 83 67 93 68 66 73 50 44 84 77 49 44 67 72 52 48 44 76 80 50 55 59

Hexadecimal => 45 53 43 5D 44 42 49 32 2C 54 4D 31 2C 43 48 34 30 2C 4C 50 32 37 3B

3.6.1.2 Example of parameters for V5/V4 embossers and Duxbury

Example:

ASCII => [ESC]DBT0,TD0,PN0,LS50,DP2,BI2,TM1,CH40,LP27;

Decimal => 27 68 66 84 48 44 84 68 48 44 80 78 48 44 76 83 53 48 44 68 80 50 44 66 73 50 44 84 77 49 44 67 72 52 48 44 76 80 50 55 59

Hexadecimal => 1B 44 42 54 30 2C 54 44 30 2C 50 4E 30 2C 4C 53 35 30 2C 44 50 32 2C 42 49 32 2C 54 4D 31 2C 43 48 34 30 2C 4C 50 32 37 3B

3.6.2 Binding Margin – BI

Binding Margin is the number of characters the embosser should leave empty on the line, to the right or to the left. This is convenient if the document is to be placed in a binder. If the

paper holes are to be on the left side, the empty characters are placed to the left, before the embossed characters, and on the right side otherwise. Make sure that BI + CH fits the paper chosen.

The current setting for binding margin can be changed with the parameter BI. The binding margin is always specified in number of characters. Valid values are 0 – 10. CH parameter must be specified if using BI.

Example:

ASCII => [ESC]DBI2,CH27;

Decimal => 27 68 66 73 50 44 67 72 50 55 59

Hexadecimal => 1B 44 42 49 32 2C 43 48 32 37 3B

3.6.3 Braille Table – BT (as of 2011-11-25. Appiles to firmware 1.3.3)

To change the Braille translation table this parameter can be used. Valid values are 0 – 25 (see table 3). Value 0 will always use the Index Braille table 6-dot (MIT table) that is built in to the embosser. Value 1- 4 will be Index defined Braille tables to be determined. Values 5 to 26 are country specific tables. All other numbers makes V5/V4 use the default setting on the embosser itself.

Example:

ASCII => [ESC]DBT0;

Decimal => 27 68 66 84 48 59

Hexadecimal => 1B 44 42 54 30 3B

0	index_braille_computer_6_dots
5	american_computer_6_dots
6	american_computer_8_dots
7	english_computer_8_dots
8	english_literal_without_capital
9	ibm437_computer_6_dots
10	ibm437_computer_8_dots
11	ibm850_computer_8_dots
12	german_computer_8_dots
13	italian_literal_with_capital
14	italian_literal_without_capital
15	portuguese_literal_with_capital
16	spanish_computer_6_dots
17	spanish_computer_8_dots
18	spanish_user_defined_3_ANSI
19	spanish_user_defined_4_ASCII
20	swedish_literal_without_capital
21	polish_computer_6_dot
22	polish_literal_with_capital
23	polish_own_table_03
24	polish_own_table_04
25	dutch_computer_6_dot

Table 3: BT values.

3.6.4 Characters per Line – CH

“Characters per line” is pretty much what it sounds like, the maximum number of characters that can fit on one line. Binding Margin (see 3.6.2) is excluded.

If CH is sent to the embosser one must also include BI parameter.

Example, a character is 6 mm wide. A A4 paper is used and it is 210 mm wide. On a 210 mm wide paper it will in theory fit $210/6 = 35$ characters. If setting the Binding margin to zero (0), in “WinBraille” mode, all 35 characters will be available for embossing, if setting this parameter to 35. However, using all the width of the paper is possible, we recommend using a slightly lower value. If the paper is pulled into the printer at a small angle, it is possible that some dots will be on the edge on the paper, or even outside.

The maximum value that currently can be used is 49 for Basic-D V5/V4, 48 for Everest-D V5/V4 and BrailleBox. This is a limitation in the printer heads total movement and only applicable to the printers that can use papers that big.

Example:

ASCII => [ESC]DCH33,BI2;

Decimal => 27 68 67 72 51 51 44 66 73 50 59

Hexadecimal => 1B 44 43 48 33 33 2C 42 49 32 3B

3.6.5 Page Mode – DP

This parameter has been extended to not only include setting for single or double sided print, but also information about Z-folding and Saddle stitch binding.

Basic-D V5/V4:

DP1 = Single sided (normal horizontal printing)

DP2 = Double sided (normal horizontal printing)

DP3 = Double sided z-folding (normal horizontal printing)

DP5 = Single sided z-folding (normal horizontal printing)

DP6 = Double sided sideways z-folding (vertical printing)

DP7 = Single sided sideways z-folding (vertical printing)

Everest-D V5/V4:

DP1 = Single sided (normal horizontal printing)

DP2 = Double sided (normal horizontal printing)

DP4 = Double sided saddle stitch 4-pages (vertical printing)

DP8 = (Not in firmware 1.3.1) Single sided saddle stitch 4-pages (vertical printing)

BrailleBox V5/V4:

DP1 = Single sided (normal horizontal printing)

DP2 = Double sided (normal horizontal printing)

DP4 = Double sided saddle stitch 4-pages (vertical printing)

DP8 = Single sided saddle stitch 4-pages (vertical printing)

Paper formats in BrailleBox V5/V4 are: A4, A3, 11x17 inch, 11x11.5 inch and 11x8.5 inch.

Example:

ASCII => [ESC]DDP1;

Decimal => 27 68 68 80 49 59

Hexadecimal => 1B 44 44 50 31 3B

3.6.6 Driver Version – DV

This parameter is used by the Index Braille embosser drivers and tells which driver version the document is generated with. The format is DV[driver version text]. This parameter does not control anything in the embossers.

Example:

ASCII => [ESC]DVversion6;

Decimal => 27 68 86 118 101 114 115 105 111 110 54 59

Hexadecimal => 1B 44 56 76 65 72 73 69 6F 6E 36 3B

3.6.7 First Line Offset – FO

The setting for first line offset is changed permanently in the embosser with HELP + 1. This parameter can be used to temporary set a new first line offset for a document. The value must be equal or greater than 0. The value specified is in tenths of millimeters, i.e. the value 35 is 3.5 mm.

Example:

ASCII => [ESC]DFO35;

Decimal => 27 68 70 79 51 53 59

Hexadecimal => 1B 44 46 4F 33 35 3B

3.6.8 Graphic Dot Distance – GD

To change graphic dot distance the parameter GD can be used. Valid values are:

0 ➔ 2.0 mm

1 ➔ 2.5 mm

2 ➔ 1.6 mm

Example:

ASCII => [ESC]DGD1;

Decimal => 27 68 71 68 49 59

Hexadecimal => 1B 44 47 44 31 3B

3.6.9 Lines per page – LP

Lines per page in current Braille Cell Size ([ESC]TD)

If LP parameter is used one must include the TM parameter as well.

Example:

ASCII => [ESC]DLP27,TM1;

Decimal => 27 68 76 80 50 55 44 84 77 49 59

Hexadecimal => 1B 44 4C 50 32 37 2C 54 4D 31 3B

3.6.10 Line Spacing – LS

To change line spacing for a document the parameter LS can be used. Valid values are:

50 → 5.0 mm (single - normal)

100 → 10.0 mm (double)

This parameter can only be specified in the beginning of a document and is ignored if a row has been added to the first page. Line spacing is ignored if DP is set to DP4 and DP8.

Example:

ASCII => [ESC]DLS60;

Decimal => 27 68 76 83 54 48 59

Hexadecimal => 1B 44 4C 53 36 30 3B

3.6.11 Multiple Copies – MC

To print multiple copies of a document the parameter MC can be used. Valid values are 1 – 10000.

Example:

ASCII => [ESC]DMC5;

Decimal => 27 68 77 67 53 59

Hexadecimal => 1B 44 4D 43 35 3B

3.6.12 Multiple Impact – MI

To impact the hammers multiple times for a Braille row this parameter can be specified. Valid values are 1 – 3. The setting will be changed for all rows after this parameter is specified.

Example:

ASCII => [ESC]DMI2;

Decimal => 27 68 77 73 50 59

Hexadecimal => 1B 44 4D 49 32 3B

3.6.13 Page Number – PN

To change page number setting the parameter PN can be used. Valid values are:

0 → None

1 → Top (require top margin > 0)

2 → Top-left (require top margin > 0)

3 → Top-right (require top margin > 0)

4 → Bottom (require bottom margin > 0)

5 → Bottom left (require bottom margin > 0)

6 → Bottom right (require bottom margin > 0)

This parameter needs to be specified before the first row is added to the first page.

Example:

ASCII => [ESC]DPN1;

Decimal => 27 68 80 78 49 59

Hexadecimal => 1B 44 50 4E 31 3B

3.6.14 Braille Cell Size (formerly Text Dot Distance) – TD

To change Braille Cell Size the parameter TD can be used. Valid values are:

0 ➔ Interpoint 2.5 mm

1 ➔ Interline 2.2 mm

2 ➔ Interpoint 3.2 mm

Example:

ASCII => [ESC]DTD2;

Decimal => 27 68 84 68 50 59

Hexadecimal => 1B 44 54 44 32 3B

3.6.15 Top Margin – TM

The current setting for top margin can be changed with parameter TM. The top margin is always specified in lines. Valid values are 0 – 10.

If TM parameter is used one must include the LP parameter as well.

Example:

ASCII => [ESC]DTM2,LP25;

Decimal => 27 68 84 77 50 44 76 80 50 53 59

Hexadecimal => 1B 44 54 4D 32 2C 4C 50 32 35 3B

3.6.16 Paper Size – PA (Only for V5 embossers, firmware 2.x)

The current setting for paper size can be changed with parameter PA. The paper size is specified with a single integer value (ID), each value represents a specific predefined paper size. See below for a list of the currently predefined paper sizes that are available for BrailleBox V5. (If the paper size setting is used, then the DP, CH, LP, TM and BI parameters must also be set.)

Example (A3-tray, booklet, 32 chars/row, 28 lines/page, 1 Top margin, 2 Binding margin):

ASCII => [ESC]DPA30,DP4,CH32,LP28,TM1,BI2;

Hexadecimal => 1B 44 50 41 33 30 2C 44 50 34 2C 43 48 33 32 2C 4C 50 32 38 2C 54 4D 31 2C 42 49 32 3B

Table 2. Predefined paper sizes

Id	Description	Width (mm)	Height (mm)	Model
30	A3-Tray	297.0	420.0	BrailleBox
31	Tabloid (11X17)-Tray	279.4	431.8	BrailleBox
32	11.5X11-Tray	279.4	292.1	BrailleBox

3.7 Transparent Mode – Multiple characters

Syntax:

27 92 x y [data] 13 10

Where x and y defines the number of data bytes/characters according to the formula below:

Number of characters = $(y * 256 + x)$. Always finish each graphics line with line feed (decimal: 13 10, hexadecimal: 0d 0a).

Each byte is translated to a 2x4 dot matrix according to the table below (each table cell corresponds to a dot e.g. Bit 3 is dot 7):

Bit 0 = dot 1	Bit 4 = dot 4
Bit 1 = dot 2	Bit 5 = dot 5
Bit 2 = dot 3	Bit 6 = dot 6
Bit 3 = dot 7	Bit 7 = dot 8

3.8 Transparent Mode – One character

Same translation as for multiple characters but no calculation is needed since only one character is translated.

Syntax:

27 94 c

Where c is the character to be translated.

3.9 Letter Graphics

Letter graphics is activated / deactivated with ESC 252. Letter graphics is embossed according to the loaded letter font in the embosser. The letter font in the embosser can easily be changed with Embosser utility.

Syntax:

27 252 [data] 27 252

Where [data] is the text to be embossed. The text can include line and form feed characters to format the text.

3.10 Graphics example for V5/V4 embossers

To handle images with 0.5 mm resolution, for example 2.0 mm graphics with intercepted Braille text.

Function	Decimal sequence	Hexadecimal sequence
Enter Image Mode	27 9	1B 09
Exit Image Mode	27 10	1B 0A
Set Image Data Type	27 11 r h v l s	1B 0B r h v l s
Send Image Data (= Transparent mode)	27 92 1b hb [data]	1B 5C 1b hb [data]

Parameters:

r	Number of dot rows per image data row sent through transparent mode Escape 92.
h	Horizontal offset (in tenth of mm) between each individual row in Escape 92.
v	Vertical offset (in tenth of mm) between each individual row in Escape 92.
ls	Line spacing (in tenth of mm) after a complete Escape 92 data sequence.
lb	Low byte of number of following data bytes (see the description of the transp. mode).
hb	High byte of number of following data bytes (see the description of the transp. mode).
[data]	Array of data bytes. The number of bytes is specified by lb and hb.

Table 3

Transparent mode data is always sent with 2 mm horizontal displacement between consecutive dots.

Example:

Suppose we want to send the following image to the embosser:

(● = embossed dot, ○ = unembossed space)

	Col. 1 0.0 mm	Col. 2 1.5 mm	Col. 3 3.0 mm	Col. 4 4.5 mm	Col. 5 6.0 mm	Col. 6 7.5 mm	Col. 7 9.0 mm	Col. 8 10.5 mm
Line 1 0.0 mm	●	●	●	●	●	●	●	●
Line 2 1.5 mm	●	●	●	●	●	●	●	●
Line 3 3.0 mm	●	●	○	●	●	●	○	●
Line 4 4.5 mm	●	○	○	●	○	○	○	○
Line 5 6.0 mm	●	●	●	●	○	○	○	○

A strategy would be:

1. **Enter image mode**

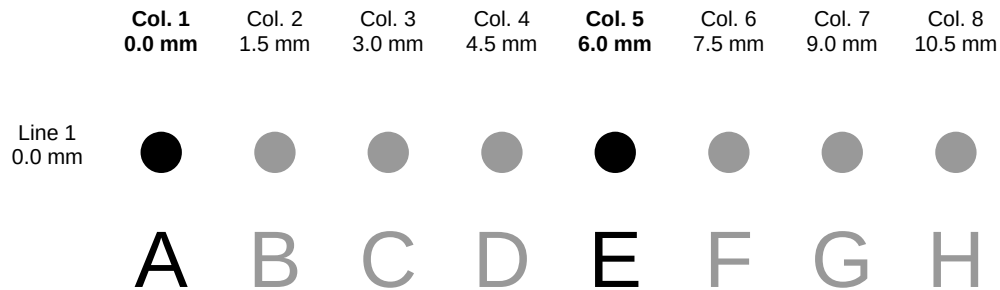
Start with the “Enter Image Mode” sequence: 27 9.

2. **Set “Image Data Type”**

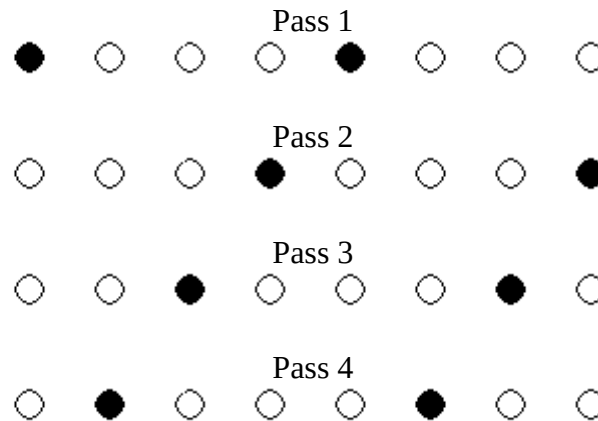
Maximum number of “rows” in a “Transparent Mode” sequence is 4, so let $r = 4$. Since the resolution is 1.5 mm between each dot and the transparent mode data is sent with 2 mm displacement, each of the four “transparent Mode” rows in the escape sequence has to be embossed with different horizontal offsets, each 0.5 mm offset from the previous row ($h = 5$), but with vertical displacement 0 ($v = 0$). This will emboss each “row” in the transparent mode sequence on the same “row” on the paper, with the first row at the leftmost position, the second with horizontal offset 0.5 mm (since $h = 5$), the third with offset 1.0 mm ($2h$), and the fourth with offset 1.5 mm ($3h$). The next line is 1.5 mm further down. We want to proceed with the next line after a transparent mode sequence of data, so let $ls = 15$. The “Image Data Type” sequence is thus: 27 11 4 5 0 15.

3. **Send the image data properly**

Send data row by row. One data byte in the Transparent Mode escape byte sequence would, for each row, emboss two dots with a displacement of 2.0 mm. Let us order the dots in the first row with the letters A to H: A B C D E F G H. Dots A and E lay in the 2.0 mm grid, as seen in the figure below. All other dots lay off-grid.



With no offset it would only be possible to emboss dots A and E (the two dots that lay on the 2.0 mm grid). The other six dots need to be embossed with an offset of 0.5, 1.0 or 1.5 mm. This is performed with four individual passes. The first pass would emboss dots A and E with no offset. The second pass would emboss dots D and H with 0.5 mm offset, the third pass would emboss dots C and F with offset 1.0 mm, and the fourth (last) pass would emboss dots B and G with offset 1.5 mm, see figure below.



The number of bytes needed totally, to emboss all dots on a line can be calculated by an equation.

B	Bytes needed
hl	Horizontal position of leftmost point
hr	Horizontal position of rightmost point
m	Size of used minimum offset
n	Number of bits in a byte

$$B = \frac{hr - hl + m}{m \cdot n}$$

Round the result up to the nearest integer.

$$B = \frac{10.5 - 0.0 + 0.5}{0.5 \cdot 8} = 2.75$$

Rounded up to nearest integer, we need three bytes totally to emboss all eight dots from each line, one for dots A, B and C, one for D, E and F, and one for G and H.

The “Transparent Mode” sequence, embosses dots as if they were ordered as the dots in an 8-dot Braille cell:

1	4
2	5
3	6
7	8

The dots are always ordered in a certain order in the byte. The order is: 8 6 5 4 7 3 2 1.

Let us place the eight dots into three 8-dot Braille cells. The dots A (at 0.0 mm) and E (at 6.0 mm) are placed in the three cells’ first row. This row will be embossed in the first pass without any offset.

	<i>Pass 1: Offset 0.0 mm</i>					
	Cell 1		Cell 2		Cell 3	
Basic placement	0.0 mm	2.0 mm	4.0 mm	6.0 mm	8.0 mm	10.0 mm
Placement including offset	0.0 mm	2.0 mm	4.0 mm	6.0 mm	8.0 mm	10.0 mm
Byte row 1, all cells	1 (A)	0	0	1 (E)	0	0

Dots D (at 4.5 mm) and H (at 10.5 mm) should be embossed with offset 0.5 mm. They should be embossed in the second pass, so let us place them in the three cells’ second row.

	<i>Pass 2: Offset 0.5 mm</i>					
	Cell 1		Cell 2		Cell 3	
Basic placement	0.0 mm	2.0 mm	4.0 mm	6.0 mm	8.0 mm	10.0 mm
Placement including offset	0.5 mm	2.5 mm	4.5 mm	6.5 mm	8.5 mm	10.5 mm
Byte row 2, all cells	0	0	1 (D)	0	0	1 (H)

Dots C (at 3.0 mm) and G (at 9.0 mm) should be embossed in the third pass with offset 1.0 mm, so place them in the third row:

	<i>Pass 3: Offset 1.0 mm</i>					
	Cell 1		Cell 2		Cell 3	
Basic placement	0.0 mm	2.0 mm	4.0 mm	6.0 mm	8.0 mm	10.0 mm
Placement including offset	1.0 mm	3.0 mm	5.0 mm	7.0 mm	9.0 mm	11.0 mm
Byte row 3, all cells	0	1 (C)	0	0	1 (G)	0

The last two dots, B (at 1.5 mm) and F (at 7.5 mm), should be embossed with offset 1.5 mm in the last pass (pass 4), so place them in the fourth row:

	<i>Pass 4: Offset 1.5 mm</i>					
	Cell 1		Cell 2		Cell 3	
Basic placement	0.0 mm	2.0 mm	4.0 mm	6.0 mm	8.0 mm	10.0 mm
Placement including offset	1.5 mm	3.5 mm	5.5 mm	7.5 mm	9.5 mm	11.5 mm
Byte row 4, all cells	1 (B)	0	0	1 (F)	0	0

The three resulting 8-dot Braille cells will be:

	Cell 1		Cell 2		Cell 3	
Byte row 1	1	0	0	1	0	0
Byte row 2	0	0	1	0	0	1
Byte row 3	0	1	0	0	1	0
Byte row 4	1	0	0	1	0	0

or, as an image:

Cell 1		Cell 2		Cell 3	
1	4	1	4	1	4
●	○	○	●	○	○
2	5	2	5	2	5
○	○	●	○	○	●
3	6	3	6	3	6
○	●	○	○	●	○
7	8	7	8	7	8
●	○	○	●	○	○

The first 8-dot Braille cell will contain dots 1, 6 and 7 (using the 8-dot Braille bit-counting method), the second cell will contain dots 2, 4 and 8, and the third cell dots 3 and 5. The three resulting bytes will be “01001001”, “10010010” and “00100100” respectively. This is 49, 92 and 24 written with hexadecimal symbols or 73, 146 and 36 with decimal numbers.

The first row in the image would thus be embossed with the escape sequence:
27 92 3 0 73 146 36.

The second row is the same as the first, so that can also be embossed with the same sequence:

27 92 3 0 73 146 36.

The third row lacks dots C and G, so the three bytes in the sequence are: “00001001”, “10010010” and “00100000”, i.e. 9, 146 and 32.

The complete escape sequence for the third row is thus: 27 92 3 0 9 146 32.

The two last rows contain only dots A to D, so these rows can be embossed with just two data bytes, row four with the values 1 and 2, and row five with values 73 and 2.

The sequences for the last two rows are: 27 92 2 0 1 2 and 27 92 2 0 73 2.

The rows have to end with line breaks, 0D 0A (hex), 13 10 (dec).

4. **Exit image mode**

End the embossing of the image and re-enter the text mode with the “Exit Image Mode” escape sequence: 27 10.

Summary: The whole sequence for this small image would be:

27 9	Activate image graphics
27 11 4 5 0 15	Set image data
27 92 3 0 73 146 36 13 10	Image data, line 1 + line feed
27 92 3 0 73 146 36 13 10	Image data, line 2 + line feed
27 92 3 0 9 146 32 13 10	Image data, line 3 + line feed
27 92 2 0 1 2 13 10	Image data, line 4 + line feed
27 92 2 0 73 2 13 10	Image data, line 5 + line feed
27 10	Deactivate image graphics

Or in hexadecimal notation:

```
1B 09
1B 0B 04 05 00 0F
1B 5C 03 00 49 92 24 0D 0A
1B 5C 03 00 49 92 24 0D 0A
1B 5C 03 00 09 92 20 0D 0A
1B 5C 02 00 01 02 0D 0A
1B 5C 02 00 49 02 0D 0A
1B 0A
```

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